Do Now · retrieval

Assembly language and addressing modes

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which primary memory is normally volatile?

Q2

What is the MAR used for?

Q3

What does an array index indicate?

Q4

For int[] a={5,8,3,7}, give a[2], then execute a[1]=a[1]+a[2]. State the new array and total.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Literal versus location — Immediate addressing embeds a literal value; direct addressing points to the memory address holding a value.

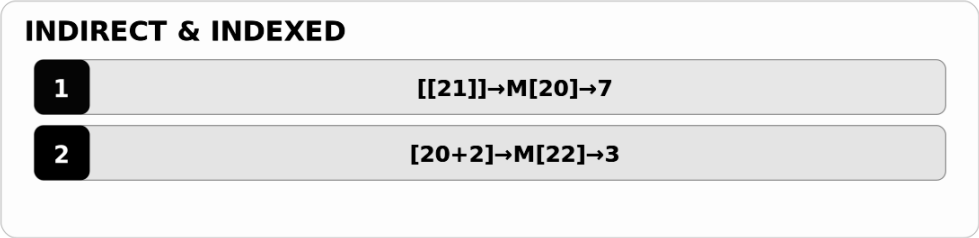
Core idea 2

Follow a pointer for indirect — M[21] contains 20, so an indirect read through address 21 accesses M[20]=7.

Core idea 3

Calculate indexed address — Base 20 plus index X=2 gives effective address 22 and value 3. These operand formats are generic examples rather than LMC syntax.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

immediate — Operand value encoded as a literal.

direct — Memory address of data specified in instruction.

indirect — Specified memory holds a further effective address.

indexed — Base address modified by index or displacement.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

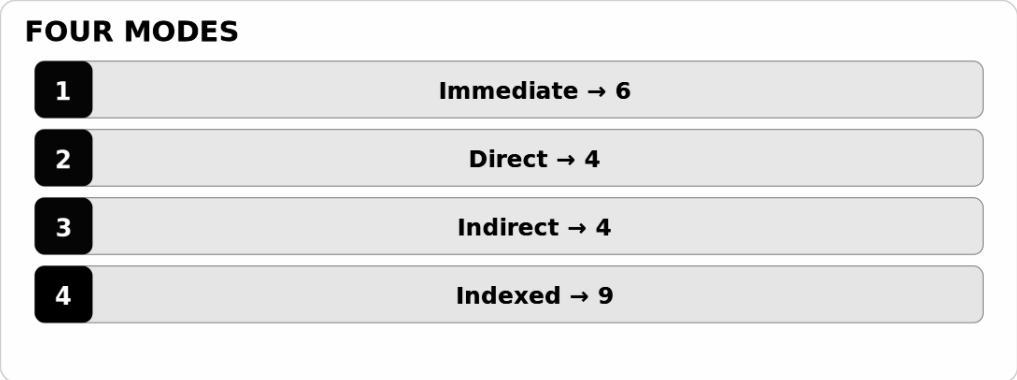
M[30]=4, M[31]=30, M[32]=9, X=2. Evaluate #6, [30], [[31]], [30+X].

Follow the worked method

1. Predict — Immediate is literal 6; direct gets 4 from memory address 30.

2. Explain — Indirect follows pointer 30 stored at 31 to value 4; indexed adds offset 2 to base 30 to read 9.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict
Immediate returns 8; direct reads M[40]=12.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define immediate, direct, indirect and indexed addressing.

Question 2

Given M[10]=3, M[11]=10, M[12]=9 and X=2, find #4, [10], [[11]] and [10+X], explaining both address calculations.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

An array of 16-bit values begins at byte address 100. Find byte address of array[3] and explain why base+3 would be wrong for byte-addressed memory.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Indirect addressing returns the pointer value itself rather than following the pointer.

Use key vocabulary from memory

1. Explain immediate in your own words.

2. Explain direct in your own words.

3. Explain indirect in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.